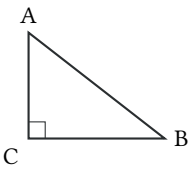
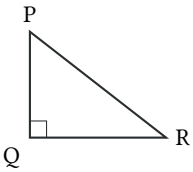
1-to’plam. *Har bir uchburchak va burchak uchun qarshi tomon, qo’shni tomon va gipotenuzani ayting.*



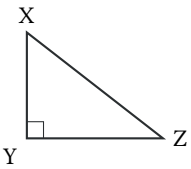
1 Burchak A uchun: . . . . . .

2 Burchak B uchun (xuddi shu uchburchak): . . . . . .



3 Burchak P uchun: . . . . . .

4 Burchak R uchun (xuddi shu uchburchak): . . . . . .



5 Burchak X uchun: . . . . . .

6 Burchak Z uchun (xuddi shu uchburchak): . . . . . .

2-to’plam. *Har bir uchburchak uchun* $\sin\theta$, $\cos\theta$, $\tan\theta$ *ni toping (kasrni soddalashtiring).*

1 Katetlar 3 va 4, gipotenuza 5. θ – uzunligi 3 bo’lgan katetga qarshi.

2 Xuddi shu uchburchak. θ – uzunligi 4 bo’lgan katetga qarshi.

3 Katetlar 5 va 12, gipotenuza 13. θ – uzunligi 5 bo’lgan katetga qarshi.

4 Xuddi shu uchburchak. θ – uzunligi 12 bo’lgan katetga qarshi.

5 . Katetlar 8 va 15, gipotenuza 17. θ – uzunligi 8 bo’lgan katetga qarshi.

6 Katetlar 7 va 24, gipotenuza 25. θ – uzunligi 24 bo’lgan katetga qarshi.

3-to’plam. *Etishmayotgan tomonni toping (yuzdan birigacha yaxlitlang).*

1 Gipotenuza = 12, burchak = 25°. Qarshi tomonni toping.

2 Gipotenuza = 18, burchak = 40°. Qo’shni tomonni toping.

3 Qo’shni = 10, burchak = 55°. Qarshi tomonni toping.

4 Qarshi = 9, burchak = 32°. Gipotenuzani toping.

5 Gipotenuza = 22, burchak = 63°. Qo’shni tomonni toping.

6 Qo’shni = 14, burchak = 47°. Gipotenuzani toping.

4-to’plam. *Etishmayotgan burchakni toping (o’ndan birigacha yaxlitlang).*

1 Qarshi = 6, gipotenuza = 14. Burchakni toping.

2 Qo’shni = 9, gipotenuza = 20. Burchakni toping.

3 Qarshi = 11, qo’shni = 15. Burchakni toping.

4 Qarshi = 18, gipotenuza = 30. Burchakni toping.

5 Qo’shni = 21, qarshi = 25 (bir xil burchakka nisbatan). Burchakni toping.

6 Qo’shni = 13, gipotenuza = 37. Burchakni toping.

5-to’plam. *Aralash amaliy masalalar (boshqacha ko’rsatilmagan, yuzdan birigacha yaxlitlang).*

1 12 fut uzunlikdagi narvon devorga 68° burchak ostida tiralgan. U devorning qancha balandligiga yetadi?

2 Pandus gorizontal 40 fut masofada 10° burchak ostida ko’tariladi. Pandusning o’zi (gipotenuza) qancha uzun?

3 Minoradan 100 m masofadagi nuqtadan minora cho’qqisiga ko’tarilish burchagi 38°. Minoraning balandligini toping.

4 Uçurtma ipi 65 m uzunlikda va yer bilan 50° burchak hosil qiladi. Uçurtma odamdan gorizontal qancha uzoqda?

5 To’g’ri burchakli uchburchakning katetlari 9 va 12. Avval gipotenuzani, so’ng 9 uzunlikdagi katetga qarshi burchakni toping (o’ndan birigacha).

6 Samolyot 6° pasayish burchagi ostida 2000 m gorizontal masofani bosib o’tadi. Balandligi necha metrga kamayadi?

6-to’plam. *Xatoni toping – buzilgan qadamni ayting, to’g’ri javobni bering.*

1 To’g’ri burchak C da, $BC = 5$, $AC = 12$, $AB = 13$. $\sin A$ ni toping.

“$\sin A = \frac{12}{13}$*, chunki 12 – uzunroq katet.”*

Xato va to’g’ri javob: .....

2 Xuddi shu uchburchak. $\cos B$ ni toping.

“Burchak A uchun qo’shni $AC = 12$ edi, shuning uchun $\cos B = \frac{12}{13}$ *ham.”*

Xato va to’g’ri javob: .....

3 $\sin 30^\circ$ ni toping.

“Kalkulyatorga 30 *ni kiritib,* $\sin 30 \approx -0.988$ *chiqdi.”*

Xato va to’g’ri javob: .....

4 $\sin\theta = 0.6$ berilgan, θ ni toping.

“$\sin\theta = 0.6$*, shuning uchun* $\theta = 0.6^\circ$*.”*

Xato va to’g’ri javob: .....

5 Gipotenuza 10, bir katet 6. Boshqa katetni toping.

“$10^2 + 6^2 =$*boshqa katet*^{2}*, demak* $\sqrt{136} \approx 11.66$*.”*

Xato va to’g’ri javob: .....

7-to’plam. *Bonus / qiyin.*

1 $\sin 23^\circ \approx 0.3907$ berilgan, $\cos 67^\circ$ ni qaytadan hisoblamay toping (qo’shimcha burchak ayniyatidan foydalaning).

2 To’g’ri burchakli uchburchakning o’tkir burchaklaridan biri 34°. Ikkinchi o’tkir burchakni toping va $\sin 34^\circ$ bilan o’sha burchak kosinusi orasidagi bog’lanishni ayting.

3 Gipotenuza 26, o’tkir burchaklardan biri 42°. Ikkala katetni toping (yuzdan birigacha).

4 20 futlik narvon devorga 16 fut balandlikda yetib boradi. Toping: (a) narvonning yer bilan hosil qilgan burchagi, (b) devor bilan hosil qilgan burchagi, (c) ikkala burchak qo’shimcha ekanini tasdiqlang.

5 Samolyot 15° doimiy burchak ostida 5000 m uchish yo’li bo’ylab ko’tariladi. Bosib o’tilgan gorizontal masofa va erishilgan balandlikni toping (har birini yuzdan birigacha).

Javoblar

To’plam	#	Javob
1-to’plam	1	Qarshi = BC, qo’shni = AC, gipotenuza = AB
	2	Qarshi = AC, qo’shni = BC, gipotenuza = AB
	3	Qarshi = QR, qo’shni = PQ, gipotenuza = PR
	4	Qarshi = PQ, qo’shni = QR, gipotenuza = PR
	5	Qarshi = YZ, qo’shni = XY, gipotenuza = XZ
	6	Qarshi = XY, qo’shni = YZ, gipotenuza = XZ
2-to’plam	1	$\sin = \frac{3}{5}$, $\cos = \frac{4}{5}$, $\tan = \frac{3}{4}$
	2	$\sin = \frac{4}{5}$, $\cos = \frac{3}{5}$, $\tan = \frac{4}{3}$
	3	$\sin = \frac{5}{13}$, $\cos = \frac{12}{13}$, $\tan = \frac{5}{12}$
	4	$\sin = \frac{12}{13}$, $\cos = \frac{5}{13}$, $\tan = \frac{12}{5}$
	5	$\sin = \frac{8}{17}$, $\cos = \frac{15}{17}$, $\tan = \frac{8}{15}$
	6	$\sin = \frac{24}{25}$, $\cos = \frac{7}{25}$, $\tan = \frac{24}{7}$
3-to’plam	1	5.07
	2	13.79
	3	14.28
	4	16.98
	5	9.99
	6	20.53
4-to’plam	1	25.4°
	2	63.3°
	3	36.3°
	4	36.9°
	5	50.0°
	6	69.4°
5-to’plam	1	$\approx$ 11.13 ft
	2	$\approx$ 40.62 ft
	3	$\approx$ 78.13 m
	4	$\approx$ 41.78 m
	5	Gipotenuza = 15; burchak $\approx 36.87^\circ$
	6	$\approx$ 210.21 m
6-to’plam	1	Qarshi/qo’shni noto’g’ri aniqlangan. To’g’ri: $\sin A = \frac{5}{13}$.
	2	Burchak almashganda tomonlar qayta aniqlanmagan. To’g’ri: $\cos B = \frac{5}{13}$.
	3	Kalkulyator radian rejimida edi. To’g’ri: $\sin 30^\circ = 0.5$.
	4	Nisbat burchak deb olingan. To’g’ri: $\theta = \arcsin(0.6) \approx 36.9^\circ$.
	5	Gipotenuza katet kabi qo’shilgan. To’g’ri: boshqa katet $= \sqrt{10^2 - 6^2} = 8$.
7-to’plam	1	$\cos 67^\circ = \sin 23^\circ \approx 0.3907$
	2	Boshqa burchak = 56°; $\cos 56^\circ = \sin 34^\circ \approx 0.5592$
	3	Qarshi $\approx$ 17.40, qo’shni $\approx$ 19.32
	4	(a) $\approx 53.13^\circ$, (b) $\approx 36.87^\circ$, (c) $53.13^\circ + 36.87^\circ = 90^\circ$ ☒
	5	Gorizontal $\approx$ 4829.63 m; balandlik $\approx$ 1294.10 m

Video yechimlar va qo’shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

YouTube

QR kod

Telegram